Searching Legislative Design Space for Compromise and Productivity

A Computational Framework for Comparing Congress-Style Institutional Designs

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Legislatures need enough productivity to act, but productivity alone can also mean volatility, agenda capture, or weakly supported lawmaking. This paper presents a reproducible simulation framework for searching legislative design space rather than defending any single reform. The model compares Congress-style institutional bundles under shared synthetic worlds: affirmative majority and supermajority rules, a stylized current-system benchmark, default enactment, scarce challenge rights, cross-bloc agenda gates, adaptive tracks, proposal accountability, active-law review, policy tournaments, citizen review, mediation, and lobbying safeguards.

The contribution is methodological. The simulator separates productivity from compromise, public alignment, low-support passage, proposer advantage, concentrated harm, lobbying capture, and reversibility. A canonical v21-paper campaign generates all reported tables and figures from one CSV artifact and reports sensitivity intervals across assumption cases. Results show that open default enactment is highly productive but risky; strong gates are safer but often blunt; and mixed bundles such as mediation plus challenge, policy tournaments, public review, adaptive tracks, and strategy-aware bundles occupy different points on the productivity/compromise frontier. The paper argues for mechanism search: legislative designs should be compared as portfolios of agenda access, contestation rights, content revision, public information, and correction mechanisms.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms**; • **Computing methodologies** → **Modeling and simulation**; • **Applied computing** → *Law*.

Additional Key Words and Phrases: legislative institutions, institutional design, computational simulation, agenda control, compromise, collective decision-making, voting systems

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1 Introduction

The motivating question is not whether one unusual voting rule should replace Congress. It is whether systematic design search can help identify legislative structures that better combine productivity, compromise, and representative responsiveness. A legislature can be unified without being democratic; authoritarian systems can appear cohesive because disagreement is suppressed. A useful representative system should instead help actors with competing interests reach compromises connected to public support, public benefit, and the burdens imposed on affected groups.

Default enactment, where a proposal passes unless a blocking coalition stops it, is one motivating example. It is useful because it reverses the usual burden of coalition formation. But it is only one design in the search space. The broader object is the mechanism bundle around any final yes-or-no vote: who may propose, how agenda access is rationed, whether information is improved before voting, whether opponents have scarce challenge rights, whether bill content can move toward compromise, whether concentrated losers receive protection, and whether bad laws can be corrected later.

This paper therefore makes a modest claim. It does not validate a new constitution and it does not forecast the U.S. Congress. It contributes an executable framework for comparing legislative mechanisms under common assumptions,

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with explicit metrics and reproducible artifacts. The current simulations are best read as comparative stress tests: they reveal when a rule produces output, when it produces compromise, and when those two diverge.

2 Related Work

The model draws on individual-level simulation, spatial voting, and institutional theory. Agent-based and computational political models are useful when heterogeneous actors interact under rules that create aggregate outcomes [de Marchi and Page 2014; Epstein 2006; Gilbert and Troitzsch 2005]. Spatial voting motivates one-dimensional ideal points and status-quo-relative utility [Downs 1957]. Legislative bargaining and agenda-setter models motivate explicit proposer identity and proposer gain [Baron and Ferejohn 1989; Romer and Rosenthal 1978]. Pivotal-politics and veto-player theories motivate comparing simple majority, supermajority, bicameral, and executive-veto systems [Krehbiel 1998; Tsebelis 2002]. Committee and agenda-control theory motivates treating floor denial, referral, information, and amendment control as outcomes rather than missing observations [Cox and McCubbins 2005; Gilligan and Krehbiel 1987; Krehbiel 1991; Shepsle 1979; Shepsle and Weingast 1987; Weingast and Marshall 1988].

Several tested mechanisms come from adjacent literatures. Lobbying is modeled as access, effort, information, public pressure, and participation bias rather than only direct vote buying [Austen-Smith 1993, 1995; de Figueiredo and Richter 2014; Gilens and Page 2014; Hall and Deardorff 2006; Hall and Wayman 1990]. Citizen-panel variants draw on deliberative polling and mini-public practice [Fishkin 2009; Organisation for Economic Co-operation and Development 2020]. Agenda-credit variants echo quadratic-voting and storable-vote work on expressing intensity under scarcity [Casella 2005; Lalley and Weyl 2018]. Reversibility variants draw on temporary legislation and sunset review [Gersen 2007; Ranchordas 2014]. Policy tournaments draw on Condorcet and agenda-manipulation problems [McKelvey 1976; Tideman 1987]. Model documentation follows ODD and ODD+D conventions [Grimm et al. 2010; Müller et al. 2013].

Default-effect rules have real analogues, although usually in bounded domains. WTO dispute-settlement reports are adopted unless there is appeal or consensus not to adopt [World Trade Organization 1994]. EU fiscal-governance enforcement has used reverse qualified-majority procedures [European Parliament and Council of the European Union 2011]. Tacit-acceptance and silence procedures allow decisions to take effect unless timely objection occurs [Churchill and Ulfstein 2000]. U.S. fast-track and disapproval structures similarly shift procedural burdens in limited contexts [Davis 2024; United States Congress 1996]. These analogues suggest the same design lesson as the simulator: burden-shifting rules need guardrails around proposal access, objection, information, and correction.

3 Simulator

Each run creates a synthetic legislature, bill stream, status quo, and institutional process. Legislators have ideal points in $[-1, 1]$, party labels, district preferences, and heterogeneous sensitivity to party pressure, constituency pressure, lobbying, reputation, and compromise. Bills have a proposer, policy position, public support, generated public benefit, lobby pressure, salience, private gain, issue domain, public-benefit uncertainty, affected-group support, and concentrated harm. Enacted bills update the status quo, so votes evaluate policy movement rather than isolated proposals.

The implementation separates voting rules from legislative processes. Voting rules interpret yes/no totals. Legislative processes wrap a bill in stages such as proposal access, committee information, committee gatekeeping, bicameralism, presidential veto, challenge vouchers, adaptive routing, sunset review, earned credits, proposal bonds, explicit budgeted lobbying, distributional-harm review, mediation, competing alternatives, citizen-panel review, public objection, and law registry review. This modularity is central: the same threshold behaves differently when proposals are open, filtered, challenged, revised, or made reversible.

The stylized current-system benchmark uses House majority passage, a Senate 60 percent threshold, presidential veto, two-thirds override, and an agenda-access gate. The gate prevents the benchmark from treating every introduced bill as a floor bill; it schedules bills using observable proxies: public support, salience, policy stability, capture risk, concentrated harm, and uncertainty. It does not observe generated public benefit directly. This makes the conventional baseline pass the current calibration screen without claiming it is a fitted model of Congress.

4 Metrics and Campaign

The simulator reports a dashboard rather than a single welfare score. Higher-is-better metrics include productivity, enacted support, generated welfare, compromise, public alignment, legitimacy, anti-lobbying success, citizen certification, district alignment, and welfare per submitted bill. Lower-is-better metrics include gridlock, low-support passage, popular failure, lobbying capture, public-preference distortion, minority harm, concentrated-harm passage, proposer gain, low-support active-law share, and false-negative default passage. Floor consideration, access denial, committee rejection, challenge use, amendment use, compensation use, reversal, attention spending, objection-window use, and policy shift are diagnostic: high values can mean useful activity or institutional strain depending on context.

For display, the paper uses a directional score. Representative quality averages generated welfare, enacted support, compromise, public alignment, and legitimacy. Risk control inverts low-support passage, minority harm, lobbying capture, public-preference distortion, concentrated-harm passage, proposer gain, and policy shift. The directional score averages productivity, representative quality, and risk control. It is a reading aid, not an optimization target.

All main results come from `simulation-campaign-v21-paper`. The campaign combines broad assumption cases, weighted party-system sensitivity cases, and a stylized rising-contention timeline in one CSV. Broad cases vary polarization, party loyalty, compromise culture, lobbying pressure, constituency pressure, party count, proposal pressure, capture pressure, and public appetite for anti-lobbying reform. Party-system cases compare two-party, two-major-plus-minor, fragmented multiparty, and dominant-party profiles. Timeline cases increase polarization, party loyalty, lobbying, and proposal pressure while reducing compromise culture and constituency responsiveness.

Calibration is a screening step, not a parameter fit. `make calibrate` compares conventional baselines with broad benchmark ranges derived from Voteview, Congress.gov, govinfo, Comparative Agendas Project, ParlGov, lobbying-disclosure records, Center for Effective Lawmaking measures, and V-Dem indicators [Comparative Agendas Project 2026; Döring and Manow 2024; Lewis et al. 2021; Library of Congress 2026; United States Government Publishing Office 2020; United States Senate Office of Public Records 2026; V-Dem Institute 2026; Volden and Wiseman 2014]. These are proxy checks for gross plausibility, not validation against raw institutional data. The current screen passes all tracked checks, including current-system enactment rate and floor load.

5 Results

Table 1 compares selected systems across broad assumption cases. The current-system benchmark is conservative: it has low productivity and low low-support passage. In this synthetic model, current-system-style gatekeeping can also protect generated welfare by allowing few bills through; that is not a dominance result because the same gate imposes a severe productivity cost. Open default enactment is the throughput extreme, but it also has much higher low-support passage and lower risk control. Mixed systems occupy the more interesting middle ground. Challenge vouchers, mediation plus challenge, citizen/public review, adaptive tracks, public objection, affected-group sponsorship, policy tournaments, and strategy-aware bundles each reduce different risks at different productivity costs.

Table 1. Canonical v21-paper scenario averages across broad assumption cases. Entries are mean±95% sensitivity interval across assumption cases, not seed uncertainty.

Label	Scenario	Directional ↑	Prod. ↑	Comp. ↑	Enacted/run	Low-sup. ↓	Risk ctrl. ↑
Current	**Current U.S.-style system**	**0.546±0.007**	**0.047±0.015**	**0.374±0.006**	**4.34±1.64**	**0.000±0.000**	**0.928±0.006**
SM	Unicameral simple majority	0.571±0.013	0.265±0.037	0.233±0.006	26.28±8.14	0.000±0.000	0.880±0.007
S60	Unicameral 60 percent passage	0.548±0.011	0.089±0.028	0.356±0.007	8.46±3.04	0.000±0.000	0.914±0.009
Bic	Bicameral simple majority	0.556±0.012	0.174±0.034	0.279±0.006	16.80±5.26	0.000±0.000	0.899±0.007
Veto	Bicameral majority + presidential veto	0.543±0.010	0.112±0.025	0.302±0.006	10.79±3.53	0.000±0.000	0.907±0.007
DP	Default pass unless 2/3 block	0.650±0.007	0.841±0.005	0.110±0.011	86.90±27.52	0.449±0.018	0.642±0.012
Chal.	Default pass + challenge vouchers	0.616±0.014	0.672±0.071	0.129±0.018	78.07±32.67	0.391±0.046	0.702±0.031
MedC	Default pass + multi-round mediation + challenge	0.671±0.010	0.785±0.043	0.217±0.008	86.35±32.23	0.301±0.044	0.731±0.017
Panel	Default pass + constituent public will + citizen panel	0.593±0.011	0.405±0.024	0.180±0.010	41.33±12.73	0.248±0.026	0.827±0.007
Adapt	Default pass + adaptive tracks strict	0.574±0.012	0.283±0.032	0.224±0.007	28.16±8.49	0.078±0.019	0.872±0.006
Pair	Default pass + pairwise policy tournament	0.710±0.009	0.550±0.012	0.491±0.022	56.36±17.50	0.042±0.028	0.930±0.007
Aff.	Default pass + affected-group sponsor gate	0.565±0.013	0.192±0.037	0.274±0.009	18.66±5.84	0.153±0.036	0.894±0.006
Obj.	Default pass + public objection window	0.596±0.012	0.447±0.027	0.176±0.009	45.30±13.81	0.269±0.023	0.813±0.004
Law	Default pass + law registry	0.658±0.007	0.867±0.004	0.136±0.012	89.63±28.46	0.398±0.017	0.634±0.012
Cost	Default pass + lobby-surcharge proposal costs	0.559±0.006	0.505±0.021	0.076±0.010	51.63±16.42	0.459±0.024	0.702±0.018
Strat	Default pass + adaptive strategy bundle	0.720±0.005	0.966±0.008	0.213±0.006	99.23±31.05	0.378±0.030	0.701±0.007

Note: All table entries come from `simulation-campaign-v21-paper.csv`. Directional, productivity, compromise, low-support, and risk-control values are normalized scores or rates. *Enacted/run* is an absolute institutional-load count. Red marks the stylized current-system benchmark.

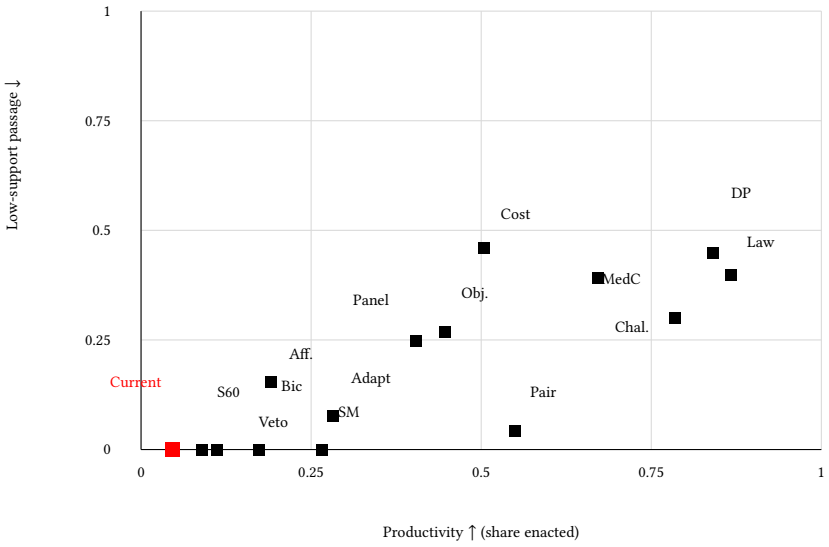

Fig. 1. Productivity versus low-support passage in the canonical v21-paper campaign. The current-system benchmark is red and marked with a larger square. Productive systems in the lower-right region are preferable under these two metrics; open default enactment moves right but also upward.

Three patterns recur in this exploratory campaign. First, passage thresholds alone are a weak design language. Supermajority and current-system baselines suppress low-support enactment, but they also suppress throughput. Open default enactment solves throughput but creates risk. Second, agenda and content mechanisms matter. Policy tournaments sharply reduce proposer agenda advantage; mediation reduces proposer gain and policy movement; citizen/public review and public objection reduce weak-support enactment; and affected-group gates improve legitimacy

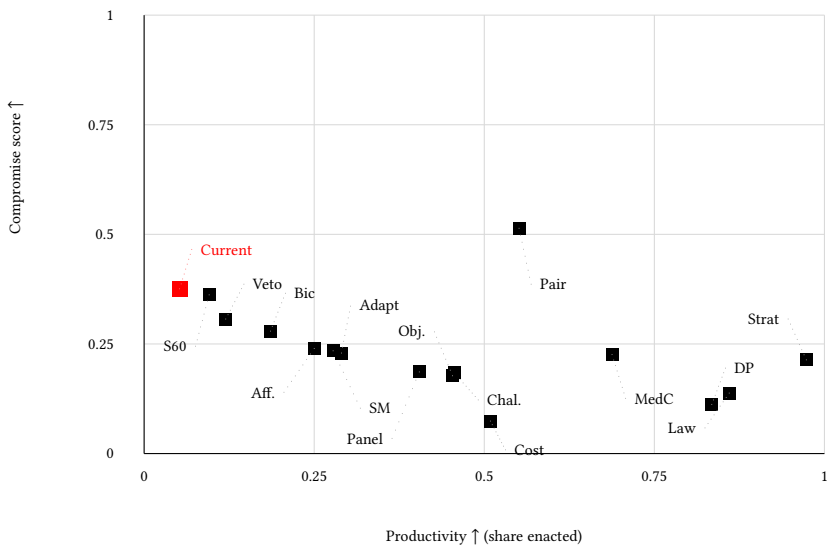

Fig. 2. Compromise versus productivity under weighted party-system sensitivity cases from the same v21-paper campaign. The plot makes the main target visible: systems should move right without collapsing downward.

at substantial access cost. Third, reversibility is a distinct tradeoff. Law-registry review reduces the stock of weak active laws but can increase status-quo volatility because correction itself changes policy.

The best current reading is therefore not "default pass is best" or "one guardrail wins." The more plausible institutional pattern is a portfolio: preserve a fast lane for low-risk proposals, require stronger consent or revision for risky proposals, allow scarce contestation rights, make proposers accountable for low-quality bills, expose organized-interest pressure, compare alternatives before binary ratification, and review enacted laws when uncertainty is high.

6 Limitations

The model is deliberately stylized. It uses one policy dimension, generated legislators, generated public support, generated public benefit, simplified issue domains, and abstract lobbying units. Public benefit is internal to the generator, so welfare comparisons are conditional on model assumptions. This creates a circularity risk: systems that screen extreme, uncertain, or lobby-heavy bills may look better partly because the generator encodes those features as lower expected public value. The paper therefore treats welfare as a shared synthetic signal, not as an empirical estimate of social welfare.

The calibration layer screens conventional baselines against named empirical ranges, but it does not fit raw datasets. It should be strengthened with adapters for bill histories, roll-call coalitions, topic throughput, lobbying distributions, district-level public opinion, and sponsor success before the model is used for stronger claims about real Congress. Strategic behavior also remains bounded. Proposers and lobby groups adapt, but parties, committees, coalitions, courts, agencies, media, and elections do not yet co-evolve over long horizons.

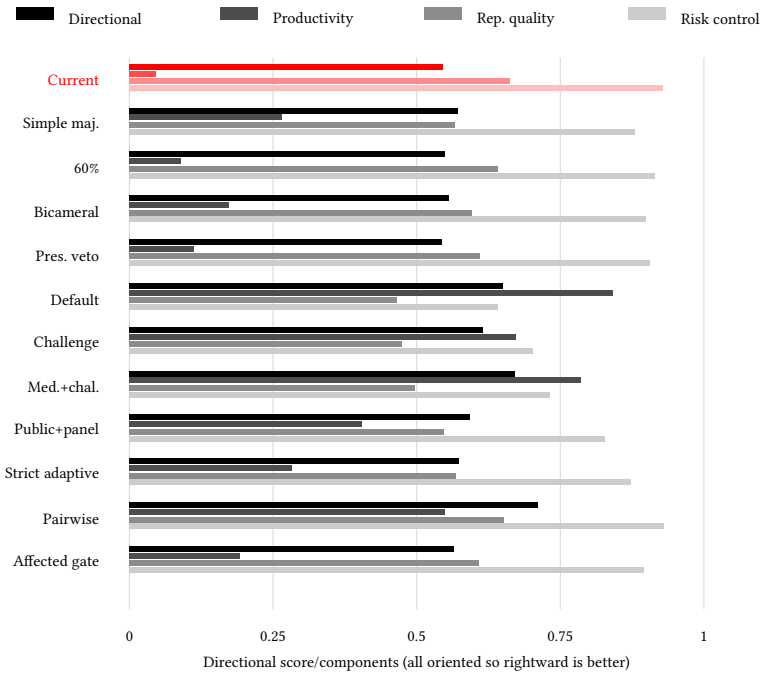

Fig. 3. Directional comparison of selected systems. Every bar is oriented so farther right is better under the stated metric interpretation. The aggregate directional score is not a normative proof; it shows how productivity, representative quality, and risk control combine.

7 Reproducibility

The submitted manuscript is anonymized for double-blind review. The reproducibility package contains source code, generated campaign reports, calibration reports, the ODD+D appendix, and scripts. The canonical results are regenerated with:

```
make campaign-v21-paper
make calibrate
make seed-robustness
make paper
make supplement-anonymous
```

The paper build regenerates the v21-paper campaign before drawing tables or figures, so the PDF is tied to the tracked campaign artifact rather than to stale CSV files. The seed-robustness target writes a separate multi-seed report for selected paper scenarios; Table 1 still reports sensitivity across assumption cases. The anonymous supplement target packages source, paper files, generated reports, and review PDFs while excluding repository metadata and local artifacts. A public repository URL will be added after review if anonymity rules permit.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. Large language model tools were used for software implementation assistance and for drafting, editing, and reorganizing manuscript text. The author

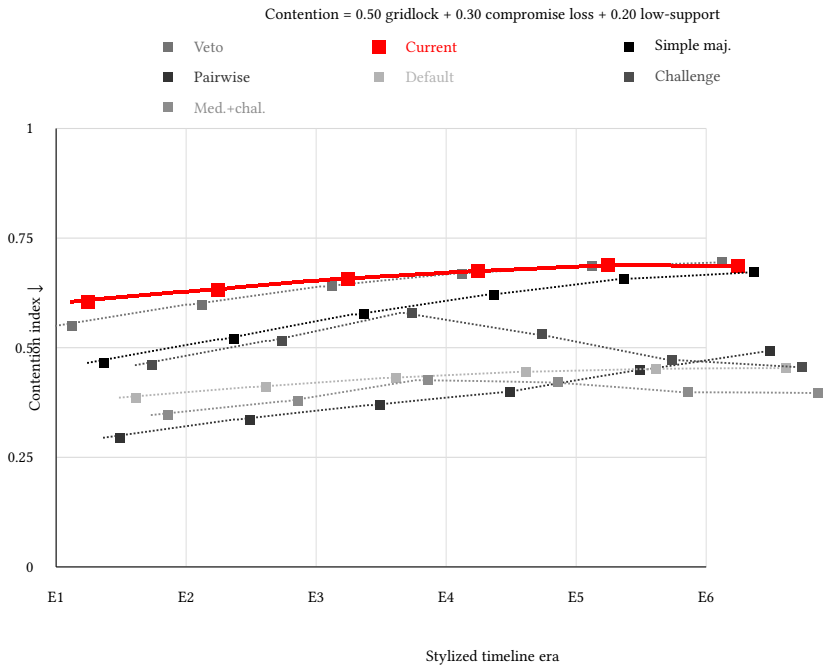

Fig. 4. Rising-contention timeline from the canonical v21-paper campaign. The index combines gridlock, compromise loss, and low-support passage. Lower is better.

reviewed and is responsible for all code, experiments, reported results, text, and citations. No large language model is listed as an author.

8 Conclusion

This paper argues for computational mechanism search as a way to study legislative design. Default enactment is one useful stress case, but the broader contribution is the ability to compare many institutional bundles under common assumptions. The current prototype suggests that compromise and productivity are most likely to improve through combinations of agenda access, contestation, mediation, alternative selection, anti-capture safeguards, affected-group protection, and reversibility rather than through any single passage threshold.

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